

# CoMn-Based Oxides as Bulk Catalyst for Rocket-Grade Hydrogen Peroxide Decomposition

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Novel bulk catalysts based on CoMn mixed oxides were prepared by a coprecipitation method for rocket-grade hydrogen peroxide decomposition. A 2 N thruster was used to evaluate and compare the catalysts in continuous and pulsed firings. The aluminum-containing mixed oxide catalyst, CoMnAl, provided the best performance, spontaneously decomposing the propellant and sustaining several thermal cycles without fractures and deactivation. Because experimental results were close to theoretical ones, this material has a good potential for peroxide decomposition in satellite and auxiliary propulsion systems.

## Nomenclature

|           |   |                         |
|-----------|---|-------------------------|
| $A_t$     | = | throat area             |
| $c^*$     | = | characteristic velocity |
| $F$       | = | thrust                  |
| $g_o$     | = | acceleration of gravity |
| $I_{sp}$  | = | specific impulse        |
| $M$       | = | mole                    |
| $\dot{m}$ | = | mass flow rate          |
| $P_c$     | = | chamber pressure        |

## I. Introduction

**M**ONOPROPELLANT propulsion systems are commonly used for orbit correction and attitude control of satellites or auxiliary functions in spacecrafts and rockets. They are simpler and present lower cost, but show lower specific impulses than liquid bipropellant propulsion systems.

Use of hydrogen peroxide ( $H_2O_2$ ) as a liquid monopropellant was studied extensively in the 1960s [1]; however, with the development of the Shell 405 catalyst (up to 36% Ir/ $Al_2O_3$ ),  $H_2O_2$  was gradually replaced by hydrazine ( $N_2H_4$ ) due to its better performance and stability [2]. Other catalysts, such as nitrides and carbides of molybdenum and tungsten [3], bimetallic catalysts (Ru-Ir/ $Al_2O_3$ ) [4], and macroscopic composites based on carbon nanofibers impregnated with iridium [5] have also been suggested for decomposition of  $N_2H_4$ . Most satellites currently in orbit around the Earth employ propulsion systems based on decomposition of  $N_2H_4$ , which is an expensive and highly toxic monopropellant. Propellant prices listed in the Aerospace Energy Standard [6] in October 2012 were 184.69/lb of  $N_2H_4$  and 0.65/lb of  $H_2O_2$  (amounts given in U.S. dollars).

Recent environmental concerns have sparked interest on storable nontoxic liquid propellants. Hydrogen peroxide is a major candidate for use as a green propellant and low-cost propulsion systems [7]. The traditional catalyst employed in this reaction is silver gauze. The use of more concentrated solutions of  $H_2O_2$  causes an increase in the adiabatic decomposition temperature of the monopropellant (632°C at 85%; 755°C at 90%; 953°C at 98%). This limits the application of pure silver or plated screens as a catalyst for long-term decomposition

of  $H_2O_2$  due to the melting point of silver (962°C) being very near the decomposition temperature [8].

Most transition metals are catalytically active in the decomposition of  $H_2O_2$ . The mechanism of decomposition follows two steps: 1) a redox reaction with electron transfer on the catalyst surface, resulting in the formation of free radicals, followed by 2) a chain reaction whereby the radicals formed react, producing  $H_2O$  and  $O_2$ . Hasan et al. [9] have made kinetic studies of the decomposition of  $H_2O_2$  in the liquid phase and claimed that 1) single or mixed oxides of transition metals in a higher oxidation state are more active in the decomposition of  $H_2O_2$  than those in lower oxidation states; 2) the rate determining step of this reaction is the electron withdrawal from the catalyst; and 3) the rate dependence on the catalyst surface area is insignificant, because the  $O_2$  hinders the reactant admission into surface pores.

The decomposition of  $H_2O_2$  for propulsion applications is extremely rapid, occurring in the vapor phase, and the thruster induction time is dependent on the number of active sites exposed to the reaction. Because this reaction is highly exothermic, a few seconds after ignition, the catalytic bed may reach a temperature sufficient to make the reaction almost entirely thermal.

Manganese dioxide is one of the most widely studied catalysts for  $H_2O_2$  decomposition. Nevertheless,  $MnO_2$  may be reduced to  $Mn_2O_3$  during the reaction with a considerable loss in catalytic activity. On the other hand, higher temperatures cause the formation of  $Mn_3O_4$ , which is as active as  $MnO_2$ . Manganese atoms with different oxidation states in the same network exchange electrons, providing the mobility necessary for the activity of the surface redox [9].

Other catalyst formulations have been proposed to replace the traditional silver catalysts [10,11], most of which are based on manganese oxides supported on alumina, silica, or titania. The authors reported that a support with a high surface area is essential for a high-activity catalyst. Rusek [10] further found that metallic silver is more catalytically active in the decomposition of  $H_2O_2$  than silver oxide and that  $Mn^{4+}$  is more active than  $Mn^{3+}$ .

Tian et al. [12] employed iridium supported on alumina (i.e., the catalyst employed in the decomposition of  $N_2H_4$ ) in the decomposition of  $H_2O_2$ . The authors observed a deactivation of this catalyst system as a function of time, caused by oxidation of the metal and adsorption of species used as stabilizers of  $H_2O_2$  on the catalyst surface. Recently, some authors have proposed the use of platinum supported on alumina as a catalyst for the decomposition of  $H_2O_2$  in microthrusters [13,14]. Despite the high cost of the active phase, the main problem with this catalytic system was the low mechanical strength of the support [15]. During the tests, catalyst grain rupture has been observed; this caused loss of the active phase, which is ejected through the nozzle. This in turn created voids in the catalyst bed, which accelerated the rate of breakage and caused instability in engine operation.

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Most published studies have suggested the use of supported catalysts for the decomposition of  $\text{H}_2\text{O}_2$ . Thus, the catalyst support needs to have high thermal and mechanical resistance, high chemical stability, and high specific surface area. The impregnation step is crucial for a good anchoring of the active phase on the support. In the case of use of manganese oxide as the active phase, the precursors most commonly used are permanganates and permanganic acid. Manganese oxides are usually formed by thermal decomposition of the precursor. A weak precursor/support interaction can result in a poor anchoring of the active phase on the support and its partial ejection during the first tests in the thruster.

Mixed oxides containing transition metals are often used as bulk catalysts in different chemical reactions. Generally, these oxides are prepared by thermal decomposition of different precursors, such as hydroxides, carbonates, nitrates, oxalates, etc. Cobalt and manganese may achieve various oxidation states and thus be effective catalysts for use in redox reactions. Kannan and Swamy [16] tested a CoAl mixed oxide as a catalyst for decomposition of nitrous oxide. The authors observed an increase in catalyst activity with increasing Co content in the catalyst system.

Bernal et al. [17] synthesized FeCo mixed oxides by the method of coprecipitation at pH 9.5. They noted that a Co/Fe molar ratio of one-half tends to generate a spinel-type phase, predominantly of  $\text{Co}^{2+}$ , when calcined at a high temperature. Samples with higher ratios have different amounts of  $\text{Co}_3\text{O}_4$  and CoO formed between 800 and 900°C.

Given the preceding and some preliminary tests already performed in our laboratory that have shown promising results, we propose the use of extruded bulk CoMn mixed oxide catalysts in the decomposition of rocket-grade  $\text{H}_2\text{O}_2$  (90%) for use as a monopropellant in satellite microthrusters. Different transition metals were used, aiming to find a material with high mechanical strength and significant surface area, even after thermal treatments at high temperatures. The materials were tested for efficient catalytic decomposition of 90%  $\text{H}_2\text{O}_2$  in a microthruster with a theoretical thrust of 2 N.

## II. Experimental

### A. Hydrogen Peroxide Concentration

Stabilized 50%  $\text{H}_2\text{O}_2$  (Solvay Chemicals) was concentrated to 90% wt/wt by a counterflow of hot dry air apparatus. The concentrated  $\text{H}_2\text{O}_2$  contained not over 400 mg/kg of nitrates and 700 mg/kg of phosphates as stabilizers. Stabilizers can reduce catalytic activity over time, however, in this work, stabilized  $\text{H}_2\text{O}_2$  with nitrates and phosphates was used for safer testing of the new catalyst.

### B. Catalyst Preparation

Four catalysts based on mixed oxides were prepared: CoMnAl, CoMnAg, and CoMnMg in a 4:1:1 ratio and CoMn in a 4:2 ratio. They were synthesized by the coprecipitation method employed by Kovanda et al. [18] using chemicals supplied by Sigma-Aldrich. Solutions containing 1 M of metallic ions prepared from appropriate nitrates and 3 M NaOH were dropped into an aqueous solution of 0.5 M  $\text{Na}_2\text{CO}_3$ , keeping the pH constant at 10. The reaction was carried out under vigorous stirring at room temperature and then kept under stirring for 1 h. The slurry was aged for 12 h before it was filtered, washed several times with distilled water, dried overnight at 60°C in air, crushed, and sieved to 100  $\mu\text{m}$ . The powder was peptized with 0.1 M  $\text{HNO}_3$  solution until attaining plastic behavior, and then it was extruded and dried at 110°C for 12 h, and calcined at 900°C for 3 h. The final product was a number of cylindrical grain catalysts approximately 2 mm in diameter and 3 mm in length.

### C. Microthruster

Performance parameters of the mixed oxide catalyst for  $\text{H}_2\text{O}_2$  decomposition were obtained by tests with a 2 N thruster, which was manufactured with modules of 316 stainless steel, including injector, nozzle, and a chamber with a catalytic bed 30 mm in length and 15 mm in diameter (Fig. 1).

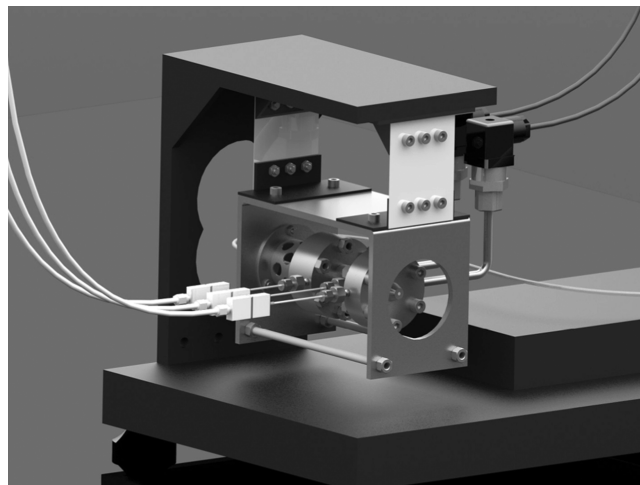


Fig. 1 Computer view of thruster and thrust balance.

The test bench included propellant and pressurizer tanks, feed line, control valves, thermocouples, pressure transducers, and a thrust balance. Hydrogen peroxide was stored in a 1 liter tank and pressurized by nitrogen, and the temperatures and pressures in the decomposition chamber were monitored, respectively, by K-type thermocouples and 0–10 bar NP430 Novus pressure transducers. The  $\text{H}_2\text{O}_2$  mass flow rate was controlled by an orifice plate previously calibrated at 2 g/s for a given differential pressure between tank and chamber. The thruster was mounted on a pendulum-type thrust balance and the thrust was measured by a compression load cell of 500 g (HBM model PW4M). A graphical user interface, written with LabView software, was used to monitor and control tests.

The Chemical Equilibrium with Applications (CEA) NASA 2004 code was used to calculate the theoretical thruster performance. A chamber pressure of 5 bar was used, and the nozzle expansion ratio was calculated as 1.45 to set the nozzle at a theoretical height of 600 m. Assuming frozen flow with 90%  $\text{H}_2\text{O}_2$ , the calculated specific impulse was 1040 m/s, the characteristic velocity was 940 m/s, and the thrust coefficient was 1.106.

## III. Results and Discussion

The CoMnMg catalyst showed high activity in the decomposition of  $\text{H}_2\text{O}_2$ , as seen in the thrust curve shown in Fig. 2a. The high activity can be justified by the specific surface area (23  $\text{m}^2/\text{g}$ ), higher than in the other catalysts, formed by the strong contraction of the material during the decomposition of the magnesium hydroxide, generating a rather porous structure with an average pore diameter of 300 Å calculated by Brunauer-Emmett-Teller method (Micromeritics analyser model 2020 Physisorption Analyzer). Nevertheless, thrust spikes occurred, probably due to the overpressure inside the pores, causing catalyst fracture and mass loss of about 20 wt%.

The addition of silver to the mixed oxides may have formed highly stable and less active silver oxides in the decomposition of  $\text{H}_2\text{O}_2$  after calcination (Fig. 2b), as already mentioned by Rusek [10].

The CoMnAl catalyst (Fig. 2d) presented the largest average thrust (~2.4 N), whereas CoMn catalyst (Fig. 2c) presented a relatively stable average thrust of approximately 1.9 N. Besides, the addition of Al to the CoMn mixed oxide increased the mechanical strength of the catalyst due to the formation of  $\alpha$ -alumina at temperatures around 900°C.

The chamber pressures showed similar profiles to the thrust curves, with higher stable pressures found with the CoMnAl catalyst. Pressure oscillations were observed with CoMnMg catalyst, causing the thrust spikes.

The CoMnAl catalyst had a low specific surface area (~6  $\text{m}^2/\text{g}$ ) due to the sinterization process at 900°C, which is above the work temperature in the decomposition chamber. This temperature was also selected to form stable spinel oxides, which are stable and resistant to rehydration. Products made at calcination temperatures at

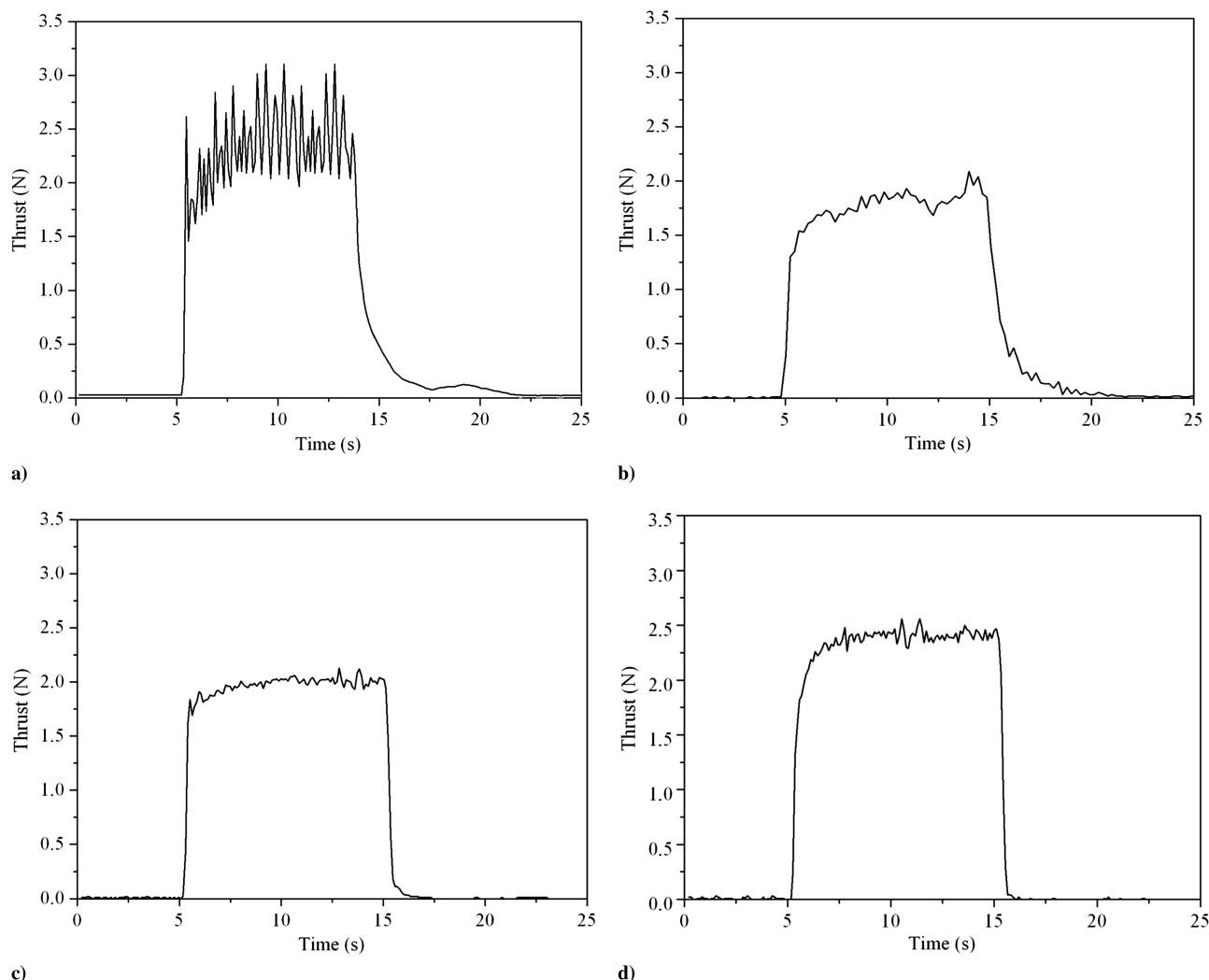


Fig. 2 Thrust curves obtained with a) CoMnMg, b) CoMnAg, c) CoMn, and d) CoMnAl catalysts.

600°C exhibit higher specific surface area (26 m<sup>2</sup>/g), but show a tendency to rehydration during H<sub>2</sub>O<sub>2</sub> decomposition. The catalyst elemental composition (52% Co, 28% O, 12% Mn, and 8% Al) was obtained by x-ray fluorescence, using a Panalytical Axios mAX spectrometer, showing 280 ppm of sodium traces.

Table 1 shows a comparison of properties of the CoMnAl bulk catalyst for H<sub>2</sub>O<sub>2</sub> and a commercial catalyst for decomposition of N<sub>2</sub>H<sub>4</sub>. The CoMnAl catalyst presents larger bed volume and mass than the Ir/Al<sub>2</sub>O<sub>3</sub> catalyst, but has a significantly lower price.

The thermogravimetric (TG) and differential thermal analysis (DTG) of the CoMnAl mixed oxide were carried out using a Thermal Analyzer Netzsch STA 449 F3 with synthetic air up to 1000°C to determine the thermal and chemical stability of the catalyst. In agreement with Kovanda et al. [18], the first peak in the DTA curve at

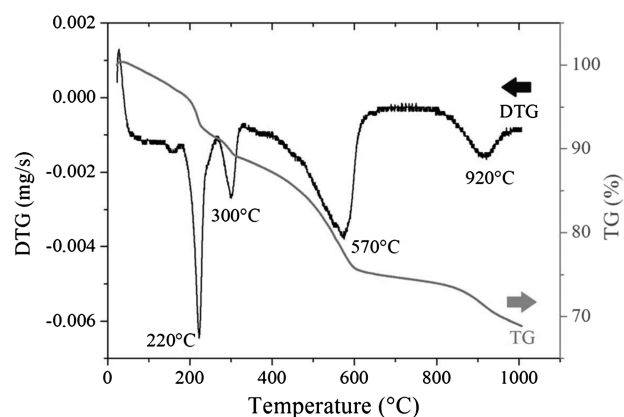


Fig. 3 DTG and TG curves for a CoMnAl catalyst heating to 1000°C in air at a rate of 10°C/min.

Table 1 Properties of catalysts for hydrazine and hydrogen peroxide decomposition

|                                          | N <sub>2</sub> H <sub>4</sub>     | H <sub>2</sub> O <sub>2</sub> |
|------------------------------------------|-----------------------------------|-------------------------------|
| Catalyst                                 | Ir/Al <sub>2</sub> O <sub>3</sub> | CoMnAl                        |
| Catalyst cost, USD/g <sup>a</sup>        | 250.00                            | 0.25                          |
| Specific surface area, m <sup>2</sup> /g | 110                               | 6                             |
| Porous volume, m <sup>3</sup> /g         | 0.10                              | 0.01                          |
| Average pore diameter, Å                 | 40                                | 80                            |
| Catalytic bed volume, mm <sup>3</sup>    | 130                               | 450                           |
| Mass of catalyst, g                      | 1.1                               | 4.6                           |

<sup>a</sup>USD, U.S. dollars; only active-phase cost, as listed by Sigma-Aldrich.

220°C (Fig. 3) was associated with the removal of the interlayered and weakly adsorbed water molecules. Dehydroxylation of the system occurred at around 300°C. The thermal decomposition of interlayered carbonate and nitrate anions began at 400°C. Over 900°C, it seems possible that some of the divalent cations were oxidized to trivalent cations, increasing the crystallinity of the system and, consequently, its chemical stability.

The thruster mass flow rate  $\dot{m}$  was determined in terms of feed pressure difference  $\Delta P$  by the equation

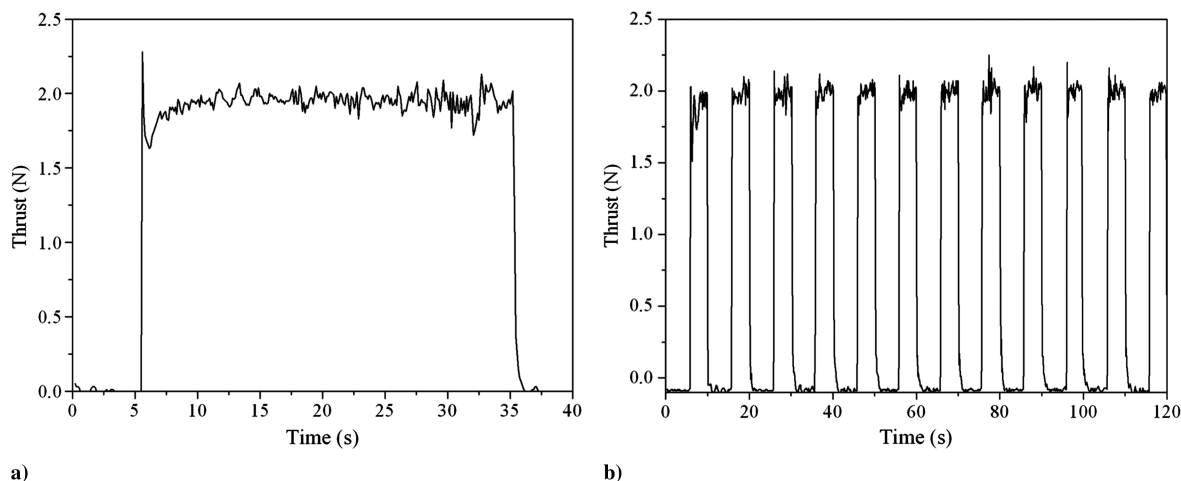


Fig. 4 Thrust curve during a) continuous 30 s test and b) 5 s on/5 s off pulse test with CoMnAl catalyst.

$$\dot{m} = 0.009\Delta P^2 + 0.613\Delta P + 0.766$$

and the specific impulse  $I_{sp}$  was calculated by

$$I_{sp} = \frac{F}{\dot{m}g_o}$$

where  $F$  is the measured thrust and  $g_o = 9.8065 \text{ m/s}^2$  is the standard acceleration of gravity. An average experimental specific impulse of 97 s was obtained. This value is relatively close to the theoretical specific impulse (106 s) calculated by CEA NASA code (2004).

The experimental characteristic velocity was obtained for a nozzle throat area of  $3.6 \text{ mm}^2$  using

$$c^* = \frac{A_t P_c}{\dot{m}}$$

Because the characteristic velocity is solely a function of temperature and gas properties in the combustion chamber, one can use this parameter to examine the chamber independently of the nozzle. The theoretical characteristic velocity was 940 m/s, whereas the experimental mean value was 822 m/s, corresponding to a characteristic velocity efficiency of 87%.

A continuous 30 s firing (Fig. 4a) and a sequence of 12, 5 s pulsed firings at intervals of 5 s (Fig. 4b) were performed to verify the stability of the CoMnAl catalyst. Chamber pressure was adjusted to yield a nominal thrust level of 2 N. In this preliminary study, there was no evidence of catalyst deactivation with respect to time, despite the use of stabilized  $\text{H}_2\text{O}_2$ . No fragmentation of the catalyst grains was observed after tests. Tests of longer duration with different duty cycles will be later carried out.

#### IV. Conclusions

Different bulk catalysts based on CoMn bulk mixed oxides were prepared by a coprecipitation method and provided various spontaneous cold starts in the decomposition of 90%  $\text{H}_2\text{O}_2$  in a 2 N microthruster. Tests were performed in an optimized catalytic bed with 30 mm length and 15 mm diameter. The CoMnAl catalyst presented high catalytic activity, despite a relatively low specific surface area of  $6 \text{ m}^2/\text{g}$ . The efficiency of this bulk catalyst can be considered excellent, because there was good repeatability, with no significant deactivation with time, and low fragmentation. The characteristic velocity efficiency was 87% and the specific impulse was 97 s, for tests at ambient conditions (600 m altitude).

New methods of precipitation of these oxides are being studied to generate a thicker texture on the substrate to increase porosity and improve access to catalytic sites. Additional tests are being performed under different operation conditions with different duty cycles and continuous firing times to evaluate catalyst stability, induction times, and fragmentation.

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#### References

- [1] Ventura, M., and Garboden, G., "Brief History of Concentrated Hydrogen Peroxide Uses," *35th AIAA Joint Propulsion Conference and Exhibit*, AIAA paper 1999-2739, 1999.
- [2] Wucherer, E. J., Cook, T., Stiefel, M., Humphries, R., and Parker, J., "Hydrazine Catalyst Production-Sustaining S-405 Technology," *35th AIAA Joint Propulsion Conference and Exhibit*, AIAA Paper 2003-5079, 2003.
- [3] Chen, X. W., Zhang, T., Ying, P. L., Zheng, M. Y., Wu, W. C., Xia, L. G., Li, T., Wang, X. D., and Li, C., "Novel Catalyst for Hydrazine Decomposition: Molybdenum Carbide Supported on  $\gamma\text{-Al}_2\text{O}_3$ ," *Chemical Communication*, Vol. 38, No. 3, 2002, pp. 288–289. doi:10.1039/b109400a
- [4] Neto, T. G. S., Gobbo-Ferreira, J., Cobo, A. J. G., and Cruz, G. M., "Ir-Ru/ $\text{Al}_2\text{O}_3$  Catalysts Used in Satellite Propulsion," *Brazilian Journal of Chemical Engineering*, Vol. 20, No. 3, 2003, pp. 273–282.
- [5] Vieira, R., Bastos-Netto, D., Ledoux, M. J., and Pham-Huu, C., "Hydrazine Decomposition over Iridium Supported on Carbon Nanofibers Composite for Space Applications: near Actual Flight Conditions Tests," *Applied Catalysis*, Vol. 279, Nos. 1–2, 2005, pp. 35–40.
- [6] "Aerospace Energy Standard Prices," Defense Logistics Agency, Fort Belvoir/VA, [http://www.energy.dla.mil/customers/standard\\_prices/Documents/Aerospace%20Energy%20FY13%20Standard%20Prices%20v3.pdf](http://www.energy.dla.mil/customers/standard_prices/Documents/Aerospace%20Energy%20FY13%20Standard%20Prices%20v3.pdf) [retrieved 15 Jan. 2014].
- [7] Wernimont, E., Ventura, M., Grubelich, M., Vaughn, M., and Escapule, W., "Low Temperature Operation of Hydrogen Peroxide Gas Generators: Verification Testing and Possible Applications," *Seventh International Energy Conversion Engineering Conference*, EC-7, AIAA Paper 2009-4617, 2009.
- [8] Kappenstein, C., Pirault-Roy, L., Guérin, M., Wahdan, T., Ali, A. A., Al-Sagheer, F. A., and Zaki, I. M., "Monopropellant Decomposition Catalysts: V. Thermal Decomposition and Reduction of Permanganates as Models for the Preparation of Supported MnOx," *Applied Catalysis*, Vol. 234, Nos. 1–2, 2002, pp. 145–153.
- [9] Hasan, M. A., Zaki, M. I., Pasupulety, L., and Kumari, K., "Promotion of the Hydrogen Peroxide Decomposition Activity of Manganese Oxide Catalysts," *Applied Catalysis*, Vol. 181, No. 1, 1999, pp. 171–179. doi:10.1016/S0926-860X(98)00430-X
- [10] Rusek, J. J., "New Decomposition Catalysts and Characterization Techniques for Rocket-Grade Hydrogen Peroxide," *Journal of Propulsion and Power*, Vol. 12, No. 3, 1996, pp. 574–579. doi:10.2514/3.24071
- [11] Sorge, A. R., Turco, M., Pilone, G., and Bagnasco, G., "Decomposition of Hydrogen Peroxide on  $\text{MnO}_2/\text{TiO}_2$  Catalysts," *Journal of Propulsion and Power*, Vol. 20, No. 6, 2004, pp. 1069–1075. doi:10.2514/1.2490

- [12] Tian, H., Zhang, T., Sun, X., Liang, D., and Lin, L., "Performance and Deactivation of Ir/ $\gamma$ -Al<sub>2</sub>O<sub>3</sub> Catalyst in the Hydrogen Peroxide Monopropellant Thruster," *Applied Catalysis*, Vol. 210, Nos. 1–2, 2001, pp. 55–62.
- [13] An, S., and Kwon, S., "Scaling and Evaluation of Pt/Al<sub>2</sub>O<sub>3</sub> Catalytic Reactor for Hydrogen Peroxide Monopropellant Thruster," *Journal of Propulsion and Power*, Vol. 25, No. 5, 2009, pp. 1041–1045. doi:10.2514/1.40822
- [14] Torre, L., Pasini, A., Romeo, L., and Cervone, A., "Performance of a Monopropellant Thruster Prototype Using Advanced Hydrogen Peroxide Catalytic Beds," *Journal of Propulsion and Power*, Vol. 25, No. 6, 2009, pp. 1291–1299. doi:10.2514/1.44354
- [15] Pasini, A., Torre, L., Romeo, L., Cervone, A., and d'Agostino, L., "Testing and Characterization of a Hydrogen Peroxide Monopropellant Thruster," *Journal of Propulsion and Power*, Vol. 24, No. 3, 2008, pp. 507–515. doi:10.2514/1.33121
- [16] Kannan, S., and Swamy, C. S., "Catalytic Decomposition of Nitrous Oxide over Calcined Cobalt Aluminum Hydrotalcites," *Catalysis Today*, Vol. 53, No. 4, 1999, pp. 725–737. doi:10.1016/S0920-5861(99)00159-5
- [17] Bernal, M. E. P., Casero, R. J. R., and Rives, V., "Preparation and Properties of Co-Fe Mixed Oxides Obtained by Calcination of Layered Double Hydroxides," *Ceramics-Silikáty*, Vol. 48, No. 4, 2004, pp. 145–154.
- [18] Kovanda, F., Rojka, T., Dobesová, J., MacHovi, V., Bezdika, P., Obalová, L., Jiráková, K., and Grygar, T. J., "Mixed Oxides Obtained from Co and Mn Containing Layered Double Hydroxides: Preparation, Characterization, and Catalytic Properties," *Journal of Solid State Chemistry*, Vol. 179, No. 3, 2006, pp. 812–823. doi:10.1016/j.jssc.2005.12.004

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